

Assessing Moderation and Multidimensionality with Mixture Models: A Reply to Broockman and Lauderdale

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Abstract

Broockman and Lauderdale (2025)—B&L hereafter—respond to our previous study (Fowler et al. 2023) by arguing that, under certain hypothetical patterns of public opinion, our statistical model inappropriately classifies some individuals as one-dimensional moderates. We acknowledge their point, but we show that none of their analyses undermine our substantive conclusions. Additional analyses reinforce our previous findings that those we classify as moderate indeed have centrist positions in the usual left-right sense. For example, when we directly account for B&L’s concern by allowing for multiple off-dimensional groups, our conclusions are largely unchanged. When we compare our model to a more complex alternative with multidimensional preferences, it performs remarkably well. The weight of the evidence suggests that most Americans have policy positions that are usefully summarized by a single ideological dimension, and among those, many have relatively moderate views.

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We are grateful to Broockman and Lauderdale (2025)—henceforth B&L—for taking up the call in our paper (Fowler et al. 2023) for renewed attention to moderates in the American electorate. B&L argue that our model over-classifies people as spatially moderate. In particular, they argue that our model performs poorly when public opinion is multidimensional and there are multiple clusters of people with idiosyncratic preferences. As a result, they conclude in their Abstract that our method “systematically overstates how many voters’ views are well-summarized by a liberal-conservative ideology.”

We re-examine the robustness of the analyses and conclusions in our original publication in light of the questions raised by B&L. Overall, these analyses indicate that the primary conclusions from our original article are robust to the concerns raised by B&L: a single ideological dimension usefully describes most, but not all, Americans’ policy views; and moderates and those whose views are not well explained by a single dimension are especially consequential for electoral selection and accountability (Fowler et al. 2023, p. 643).

The paper proceeds as follows. First, we acknowledge B&L’s main point that under certain patterns of public opinion, our Conversionian category of non-spatial respondents may be too restrictive. We then discuss the evidence presented by B&L and explain why it does not undermine our substantive conclusions. Next, we try to address B&L’s objection in the most direct possible way by allowing for multiple groups of Conversionians, and we show that this has little effect on our main findings. Then, to better assess the usefulness of our model and reliability of our results, we present additional evidence to validate our estimates. Lastly, we compare the performance of our model to an alternative with multidimensional preferences. We provide additional evidence that public opinion is usefully summarized by a one-dimensional model and that there is little additional benefit of a more complicated model.

Our Objective and B&L’s Objection

We built our statistical model to estimate ideology while accounting for the possibility that some members of the public exhibit little ideological constraint. Broockman (2016)—channeling the classic work by Converse (1964)—argued that such people were pervasive and significantly distorted the estimates of ideology arising from standard one-dimensional models (e.g., Clinton, Jackman, and Rivers 2004). Our goal was to allow respondents with an idiosyncratic mix of preferences to be pulled out of the moderate category so we could learn how many of those who appeared to be ideological centrists were, in fact, something else. This proved useful, as we removed 20.7% of respondents from the one-dimensional set. We called them Conversians and concluded that most, but not all, Americans’ policy views were usefully summarized by a single ideological dimension.

B&L’s core criticism is that our Conversian category is overly restrictive and does not “closely align . . . to the perspective articulated by Converse 1964” (p. 2, First View). They show that under certain hypothetical patterns of public opinion, our model misclassifies some individuals as Downsian whose preferences are not the result of a single ideological dimension. We acknowledge that our Conversian category may not include all of those whom Converse (1964) might classify as having relatively unconstrained views. Further, if the issue constraint (cross-issue correlations) observed in the revealed preference data used to fit the model is driven by a set of idiosyncratic (non-spatial) clusters that cannot be reduced to a meaningful single dimension, our model will nevertheless extract the best-fitting dimension from those idiosyncratic clusters and assign some fraction of voters to that dimension.

Our model estimates a dominant left-right ideological dimension, and one goal of our approach was to explore the degree to which those members of the public whose positions are not predominantly liberal or conservative can be understood as centrally located (moderate) along that left-right dimension. Our Conversian category captures one group of respondents with a set of issue-specific positions that does not neatly map onto this left-right continuum, but future readers of our paper should keep in mind that it may not capture all individuals

who do not closely fit the continuum.

B&L’s Evidence Does Not Undermine Our Key Findings

Having acknowledged B&L’s key theoretical point, we focus in the remainder of this paper on the relevance of this concern to our substantive conclusions. Because B&L see the Conversionian category as overly restrictive, they argue that we systematically overstate the number of Downsians (whom they call ideologues) and, thus, overstate the number of moderates. B&L argue that our substantive conclusions are “unreliable” (p. 1, First View) and specifically that our conclusion “that many voters who express a mix of conservative and liberal views can be inferred to support centrist policies therefore remains in doubt” (B&L, Abstract).

B&L do not definitively refute our conclusions about the prevalence of moderates, nor do they endorse an alternative measurement model against which we might compare the usefulness of our model. They show hypothetical simulations in which our model behaves in ways they find undesirable. But this neither constitutes direct evidence against our findings nor provides support for an alternative model or set of findings.

Here, we briefly summarize the evidence presented by B&L and explain why it does not undermine our substantive conclusions or the usefulness of our model. In their motivating example illustrated in their Figure 1, B&L plot simulated data with two ideological dimensions that are correlated with one another and with most of the simulated respondents clustered near the mean on both dimensions. This illustration shows that if policy preferences are two-dimensional, our method may classify some as Downsian whose views are poorly summarized by a single dimension.

However, this simulation does not undermine our main findings. First, because the two dimensions in Figure 1 of B&L are correlated by design in their simulation, most respondents are, in fact, well-summarized by a single ideological dimension. Collapsing the two dimensions into one would not lead to much loss of explanatory power. Second, although there are

respondents on the bottom-right of the picture whose views appear to be poorly summarized by a single dimension and therefore, arguably, incorrectly classified as Downsian, there are few of them. Third, even many of the people classified as Conversian are reasonably close to the mean on one or both dimensions.

In their Figure 2, B&L apply a two-dimensional model to the real data from our study in a way that makes it look like preferences are two-dimensional, as in their hypothetical data in Figure 1. In their figure, Conversians typically fall in one off-diagonal quadrant while our model classifies those in the opposite off-diagonal quadrant as Downsian. Their Figure 2 seems to suggest that a two-dimensional model would perform significantly better than our model in explaining public opinion and that our model misclassified unconstrained respondents as Downsian.

To see whether B&L’s visualization constitutes convincing evidence of multidimensionality or reveals a problem with our model, we simulated data in which the data generating process perfectly matches our mixture model. That is, we create a set of survey responses drawn from a simulated population of respondents who follow the statistical specification of our original statistical model exactly.

We simulated 1,000 respondents with 73 percent Downsians with one-dimensional ideal points distributed normally, 22 percent Conversians with item probabilities drawn from a uniform distribution between 0.1 and 0.9, and 5 percent Inattentives. We then analyzed and visualized the data exactly following B&L, who estimate a two-dimensional IRT model using the `mirt` package in the statistical software `R` and then plot the results by type.

The results of one such simulation are shown in Figure A. We have conducted many such simulations, and the results all look qualitatively similar, although Conversians can emerge in either the top-left or the bottom-right quadrant of the figure.

Figure A shows a key limitation of B&L’s evidence: Even when our mixture model is the data generating process, and even when our classifications of respondents are perfect, the method used by B&L produces a misleading visualization of the data. Just as in B&L’s

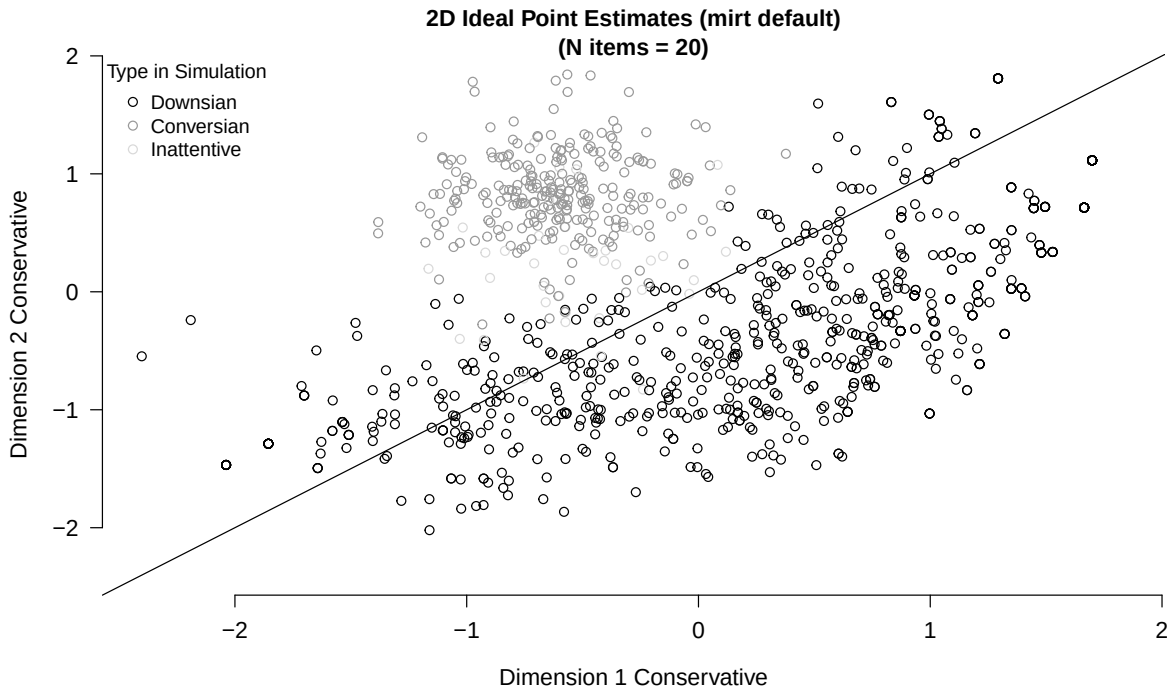


Figure A: Two-Dimensional IRT Model Estimates on Simulated Data from a One-Dimensional Data-Generating Process

Figures 1 and 2, Conversionians appear to hold off-dimensional views in one quadrant of the two-dimensional space, but a set of people classified as Downsian appear off-dimensional in the opposite quadrant. B&L’s method makes these hypothetical respondents *appear* as if they are misclassified, and it makes public opinion look two-dimensional, but because we simulated the data, we know this is not the case.

What explains the illusion? Just as an OLS regression will estimate a non-zero coefficient on a variable created by a random number generator, B&L’s two-dimensional IRT method will fit two dimensions even if the data-generating process is one-dimensional.¹ The method will find a second dimension in the response patterns of Conversionians, Inattentives, or idiosyncratic noise of Downsian survey responses.

In an additional analysis (their Table 1), B&L conduct simulations where the level of

¹Estimating a second dimension in the presence of Conversionians also distorts the interpretation and estimation of the first dimension such that individuals we know are Downsian moderates (because we simulated the data) may appear to be conservative on the first dimension and liberal on the second dimension or vice versa.

constraint across 15 issue domains varies by respondent.² In these simulations, preferences are normally distributed on each issue such that, on any given question, most respondents have preferences that are reasonably close to the mean, although in their Appendix, B&L also conduct simulations with some pure partisans.

As in Figure 1 of B&L, it is not obvious how our model should classify respondents when there is a modest amount of constraint, but it is not unreasonable to classify these respondents as Downsian. Furthermore, even some of the non-ideologues in their simulations will, by chance, happen to have preferences that are well-summarized by a single dimension, so again, it is not unreasonable for some of them to be classified as Downsian. B&L do not indicate which, if any, individuals in this simulation *should* be classified as Downsian. Importantly, because B&L’s simulations include a lot of respondents who can reasonably be classified as Downsian moderates, these exercises do not undermine our method or results.

Allowing for Multiple Conversion Groups

Our original model was purposefully parsimonious, so we are sympathetic to concerns by B&L that the parsimony might miss important features of policy views. In this section, we show how our model can be extended and generalized to account for these concerns. We note at the outset that available data limits how many parameters can be estimated by a model. So there, of course, must be some simplification to a statistical model.

The biggest concern of B&L is that there may be multiple large and different clusters of preferences in the public that don’t fit the left-right continuum. This could lead our model to misclassify some non-Downsian individuals as Downsian, and it may also contaminate our

²Some of these simulations assume that “Respondents... exhibit no *one-dimensional ideological* constraint; their views in each issue domain are entirely uncorrelated with those in all other issue domains.” We are unaware of any support in the recent empirical literature for the idea that the public holds views that are constrained within 15 different issue domains, but lack any meaningful constraint across domains. In contrast, Hare, Highton, and Jones (2024) conduct an exhaustive analysis of recent U.S. opinion data and find that “policy attitudes are organized alongside relevant core values and affective evaluations in a common, low-dimensional cognitive space. A unidimensional approximation of these belief structures often exhausts the explanatory power of vote choice models.”

estimation of the dominant Downsian dimension. An extension of our model that addresses this concern is to simply add more Conversionian categories. Scholars interested in mapping the electorate could, in principle but subject to data limitations, add more and more Conversionian categories to account for multiple patterns of policy views that are inconsistent with the spatial dimension.

Adding a second Conversionian category will induce the statistical model to find a second pattern of response probabilities that improves the model’s explanation of the observed data. So, for example, if there were one cluster of economically conservative and socially liberal libertarians and one cluster of economically liberal and socially conservative Christian democrats, the model could estimate a Conversionian group for each.

But the fact that a model could detect these additional groups does not necessarily indicate that it provides additional substantive value for describing the mass public’s views, as we show above in our Figure A and its discussion. Because the model maximizes the likelihood of the data given the parameters at its disposal, and because we believe respondents answer policy survey questions with measurement error, even if there were only single-dimensional Downsians in the population the model would still pick a range of the single-dimensional respondents to classify as Conversionian. This would lead the model to understate the number of Downsians. In other words, adding more parameters to a model is no free lunch, which is why we chose parsimony, engaged in careful validation, and created our model specification drawing on existing political science theory.

Despite these theoretical hesitations, we think it is important to examine B&L’s point that there could be a significant number of Americans in a second Conversionian group that we were misclassifying as moderates in our original paper. To do this, we extend the original model in the following way: after running our model once, we simply drop all individuals identified as Conversionian in the first estimation and then run the model a second time. This allows us to identify a second Conversionian group. One could estimate all the groups simultaneously, but we see value in the simplicity and transparency of this approach.

Figure B shows the results of this exercise using a variation on the two-dimensional visualization technique preferred by B&L. We estimate a two-dimensional IRT model for every year of the Cooperative Election Study in our data, as well as a smaller sample from 2010 that asks 133 policy questions. Separately, we estimate the model from Fowler et al. (2023) twice to allow for the possibility of a second set of parameters for Conversionians. Ovals on the graph are the smallest ovals that contain 80% of each group. We color them by category: black for Downsideans, dark grey for the first set of Conversionians, and light grey for the second set of Conversionians. The centroid of each group is marked by a corresponding letter, scaled to represent the relative number of people in that group.

In every dataset, the Downsidean group is the largest and the two dimensions are highly correlated. The Conversionian groups are smaller, and the second Conversionian group is always the smallest. The locations of the two Conversionian groups vary across data sets, evidence that the Conversionian groups and the second dimension of the visualization are strongly influenced by the particular set of questions in each survey. We discuss this further in our section on dimensionality, below. In many cases, the Conversionian groups are mostly contained within the larger Downsidean group.

The bottom-right panel of Figure B shows the same analyses for a simulated, hypothetical dataset for which 80 percent of respondents are one-dimensional Downsideans and 20 percent are from one Conversionian group. In other words, the data are consistent with our original model, and a second group of Conversionians should not be necessary to summarize preferences in this data. Interestingly, this figure looks similar to those generated from real data, although only 3 percent of simulated respondents are assigned to the second Conversionian group.

Pooling across our data sets, the mean posterior probabilities imply that 20.7 percent of all respondents are best fit to our first Conversionian group. The mean posterior probabilities from the second estimation imply that a new 10 percent of all respondents are in the second Conversionian group, leaving a majority—63.5 percent—as Downsidean. Because we find

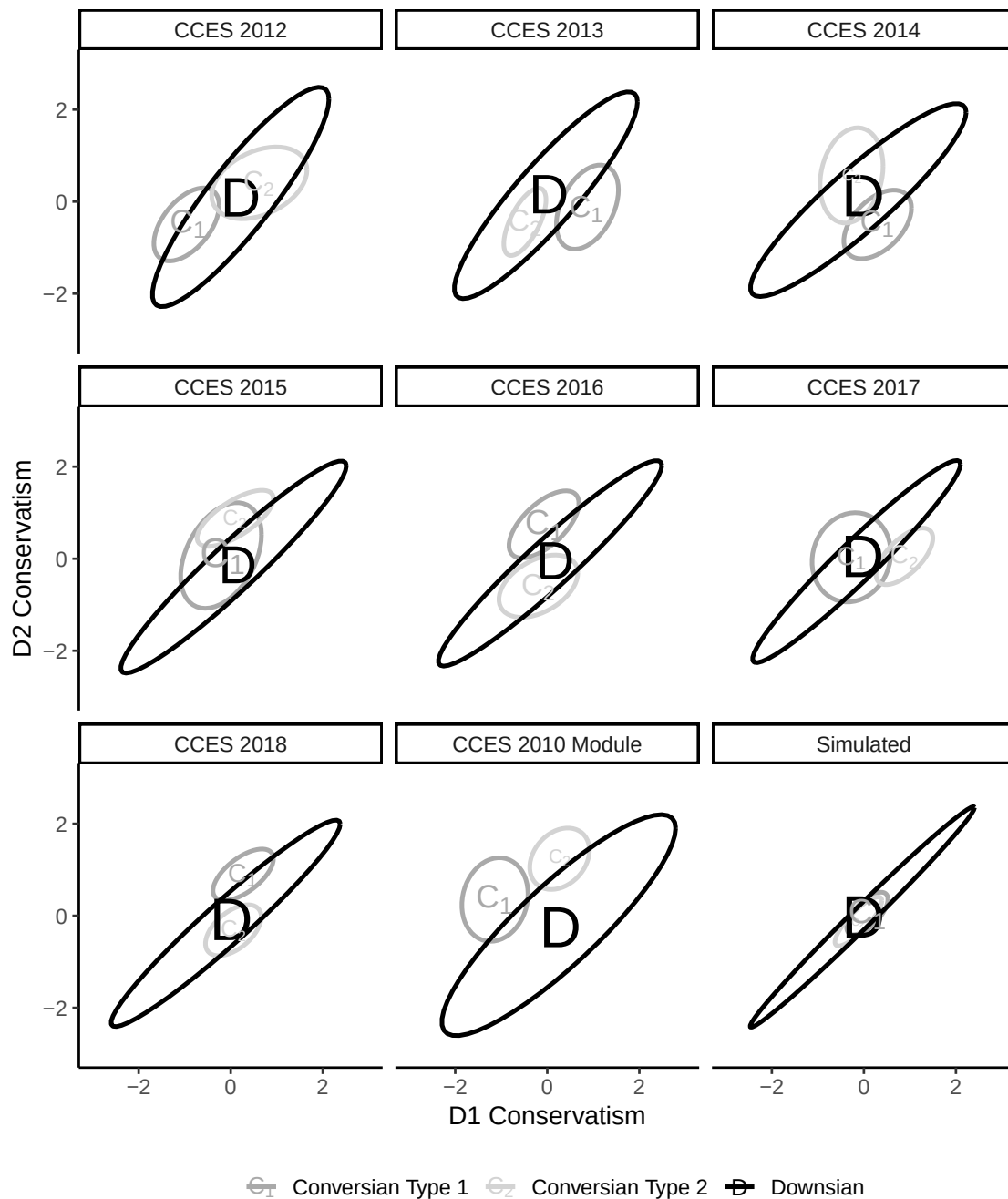


Figure B: Results with Two Conversion Groups

that adding a second Conversion category changes the number of Conversionists modestly, it suggests that adding a third, fourth, or more Conversion categories would not much further explain the data. Indeed, we have done this exercise for the 2010 module with many policy questions, and we classify only 3.2 percent of all respondents in the third Conversion group,

and only 3.1 percent in the fourth group.

In the Appendix, we reproduce every table and figure from Fowler et al. (2023) using the estimates from this extension with two groups of Conversians. As expected with the increased degrees of freedom, we estimate a smaller share of Downsians when we allow for two groups of Conversians. Every other substantive result from our original paper is virtually unchanged. Importantly, even the respondents we classify in one of the two Conversian groups tend to behave more like Downsian moderates than like ideological extremists. For example, they change their vote choices in response to the ideology and experience of congressional candidates more than do extremists.

Further Validating Our Estimates

In our original paper, we made significant efforts to assess the reliability, validity, and usefulness of our estimates. Here, we briefly review what we found, explain why it increased our confidence in the value of our model, and add one more validation that only became available to us after our paper was published.

B&L write that “the critical question underpinning the article’s conclusions is whether it is safe to assume that we can infer voters’ views on individual issues from their liberal-conservative ideology if the article’s model classified them as “Downsian” (p. 3, First View).

In our original paper, we validated this in several ways.

- In Figure 2, we examined responses to two minimum wage policy questions in the 2010 CCES module and showed that the model meaningfully differentiates among types: Downsian moderates are likely to oppose eliminating the minimum wage but also oppose increasing it to 15 dollars, extreme Downsians support the ideologically consistent positions, and inattentive respondents are disproportionately likely to give the seemingly incoherent responses of supporting both elimination and an increase.
- In Appendix C, we showed that the minimum wage example in the main body extended

to other policy views not used in estimation. Across 133 questions and 8,778 question pairs, we found that those we identify as Downsian moderates are most likely to answer pairs of questions in a moderate way.

- In Figure 3, using out-of-sample policy questions from CCES 2014 modules not used to fit the model, we found that Downsian moderates are substantially less likely than ideologues or inattentive respondents to take unpopular policy positions, consistent with genuine ideological moderation.

The subsequent substantive analyses of our paper are further consistent with the idea that the people we classify as moderate are genuinely centrist. They are more likely than others to self-identify as moderate or independent, and they are particularly willing to change the party they support in congressional elections depending on the candidates' ideologies and levels of experience.

Here, we provide an additional validation that was not available to us at the time our original paper was published. The 2023 Cooperative Election Study (Schaffner et al. 2024) contains 34 binary policy questions that we have used to estimate our model. It also asks a non-standard question about how many weeks into a pregnancy a person should be legally allowed to obtain an abortion. Respondents answered using a sliding scale from 0 to 40. This question was not an item used to estimate the mixture model, but the more continuous nature of this question allows us to assess whether the respondents classified as moderate by our model are indeed more likely to give answers in the middle of the spectrum. Figure C shows the results of this analysis. The methodology follows that of Figure 6 in our original paper. The Downsians are shown in black, Conversians are dark gray, and inattentives are light gray.

The left panel of Figure C shows how different types of respondents answered the continuous abortion question, on average. As expected, among Downsians, more liberal respondents say that abortion should be legal for more weeks. Our model, which did not use this question to classify respondents, does an excellent job predicting responses to this more continuous

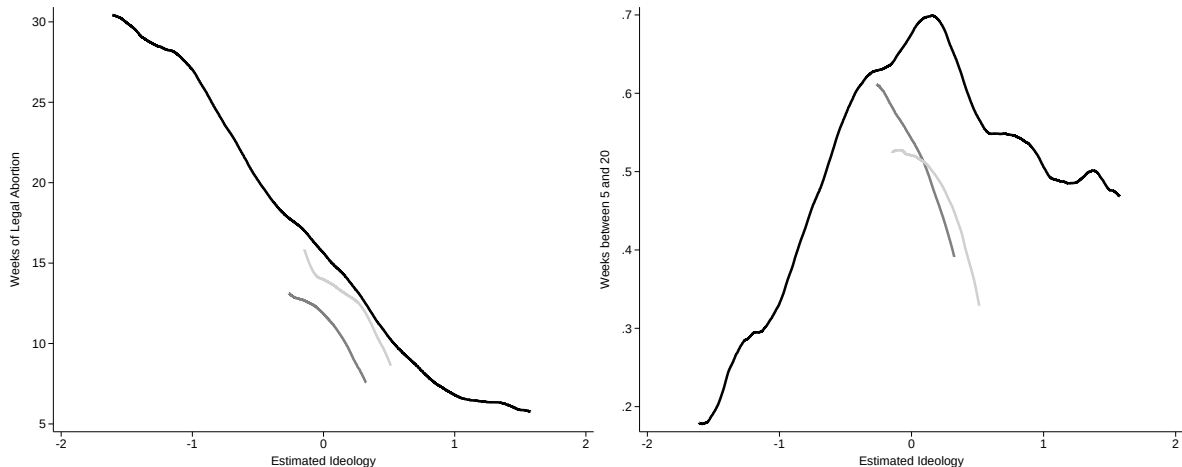


Figure C: Abortion Preferences across Types

Kernel regressions of abortion preferences by a posteriori ideology. Separate plots are shown for Downsian (black), Conversion (dark gray), and inattentive (light gray) respondents. The outcome in the left panel is the preferred number of weeks that abortion should be legal for any reason. The outcome in the right panel is an indicator for a relatively moderate response between 5 and 20 weeks.

question. For this particular question, Conversionians are, on average, more conservative than Downsian moderates.

The right panel shows which types of respondents were most likely to give a centrist response to the continuous abortion question. We code a binary indicator for whether a respondent gave a response between the 25th percentile (5 weeks) and the 75th percentile (20 weeks), and we use this as our outcome variable. Downsian moderates are the most likely group of respondents to give a centrist answer to this question, Downsian liberals are the least likely, and Conversionians, Inattentives, and Downsian conservatives are somewhere in between.

Altogether, these analyses provide strong face validity for our classifications and give us confidence that those identified as Downsian moderates by our model are meaningfully moderate in both attitudes and behavior.

Evidence on Dimensionality

Although B&L do not endorse a specific alternative model, their examples assume spatial preferences over two or more dimensions. In this section, we discuss the previous evidence and present new evidence that public opinion is very usefully summarized by a single ideological dimension. We find a limited increase in explanatory power when adding a second dimension to the model, which then suggests that adding further dimensions is unlikely to improve explanation much further.

As previously discussed, B&L’s Figure 2 appears to show evidence that preferences are multidimensional, but this is an illusion. We would expect their method to produce results like that even if the assumptions of our model are exactly right. Since B&L’s evidence is not dispositive with regard to the dimensionality of public opinion, we must turn to other evidence. In our original paper, we took the concern about higher dimensions of policy ideology seriously and implemented a two-dimensional version of our model in Appendix D, which was largely ignored by B&L. In comparing the one- and two-dimensional versions of our model, we found that the addition of a second dimension reduced the estimated share of Conversians, suggesting that some of the respondents originally classified as Conversians have preferences that can be better summarized by a two-dimensional model. Consistent with the results below, we also found that the addition of a second dimension only minimally increased the fit of the model. This suggests diminishing returns to adding further dimensions to the model.

The dimensionality of public opinion is conceptually thorny and empirically difficult to assess. There is always a tradeoff between parsimony and fit. Just as adding a variable to an OLS regression will always increase the estimated r-squared statistic, adding dimensions to a scaling model will, by construction, always increase the in-sample fit of the model to the data. Therefore, a higher fit statistic would not be sufficient evidence to conclude that a two-dimensional model is more appropriate or useful than a one-dimensional model.

Table A shows the share of survey responses correctly predicted by one- and two-dimensional

item response models run on the CCES datasets used in Fowler et al. (2023), including the 2010 module that asked 133 policy questions. It also shows the average of these fit statistics for 30 B&L simulations with two correlated dimensions, as well as a modified version of B&L’s simulation code that uses only one dimension. As expected, the two-dimensional model fits better in every case, but the difference in correct predictions is modest. The table shows that a one-dimensional model will correctly predict approximately 75 percent of respondents’ policy positions, and the addition of a second dimension increases the success of these predictions by only 1 to 3 percentage points. Even in the case of B&L’s simulations with two correlated dimensions, the addition of a second dimension only increases the percent of responses correctly predicted from 71 to 73 percent.

Table A: Analysis of Fit in One and Two Dimensions

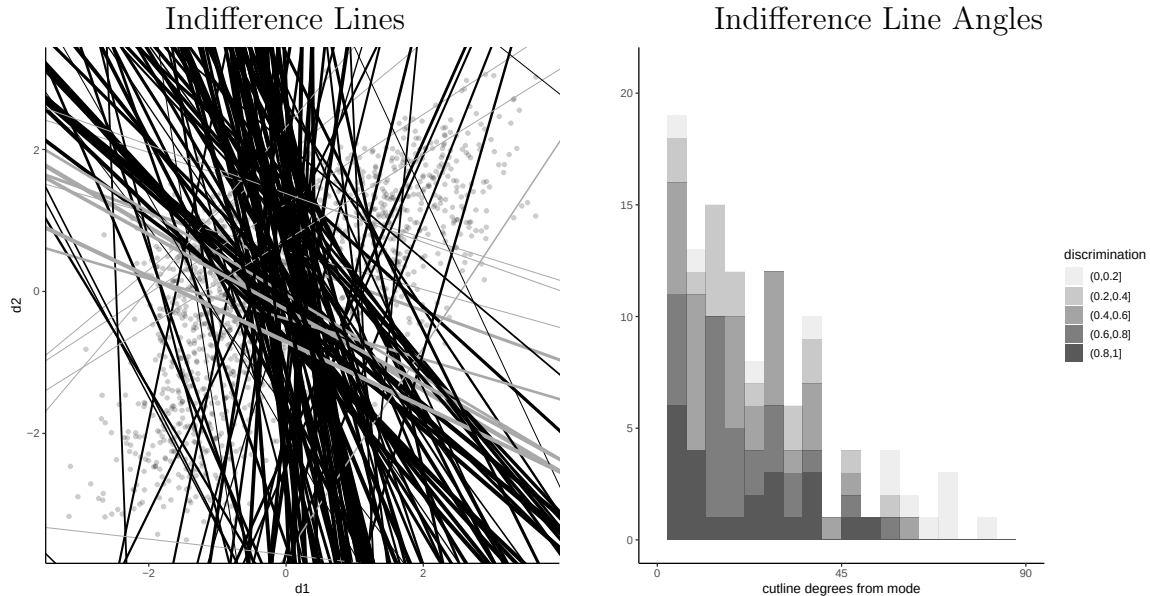
	PCP 1d	PCP 2d	Log Lik 1d	Log Lik 2d
CCES 2010 Module	.79	.80	-54355	-52235
CCES 2012	.75	.78	-609651	-594725
CCES 2013	.75	.78	-196991	-192459
CCES 2014	.74	.76	-973086	-954532
CCES 2015	.78	.80	-161623	-157275
CCES 2016	.76	.78	-987269	-967683
CCES 2017	.79	.80	-211172	-206839
CCES 2018	.78	.79	-962553	-948561
30 B&L Sims - 2D	.71	.73	-143031	-138448
30 B&L Sims - 1D	.74	.74	-134366	-134276

This analysis does not resolve the debate about whether one should prefer a one- or two-dimensional model. But it tells us that, in our data sets, a one-dimensional model fits the data reasonably well and a less parsimonious two-dimensional model only marginally improves fit. This is consistent with recent findings from Hare, Highton, and Jones (2024), who show that public opinion in the United States is mostly one-dimensional, at least in the modern era.

To further assess whether the policy views of Americans are better summarized by a one- or two-dimensional model, we can also examine how the policy questions are fit by the addition of a second dimension. We analyze the subset of 2010 CCES respondents

who answered 133 questions. Following B&L, we use the R package *mirt* to estimate their two-dimensional model, and we plot the indifference lines for each question.

Figure D: Analysis of Items in 2010 CCES Module



The results of this analysis are shown in Figure D. The points in the left panel are the ideal points of the respondents, and the lines are the estimated indifference lines for each question. The thickness of the lines is proportional to their discrimination, measured as the change in the probability of responding affirmatively to a question as an ideal point moves across the indifference line. The shading of the lines captures whether or not the angle of their slope is more than 45 degrees different from the modal angle, with items close to the mode represented in black and items far from the mode represented in gray. The right panel shows a histogram of the absolute difference between the angle of each indifference line and the modal angle.³ The bars of the histogram are shaded according to how discriminating the items are, with darker shades corresponding with more discrimination.

The cut line angles in this dataset are roughly normally distributed around -70 degrees. The right panel of Figure D shows the absolute deviation from the modal angle, which is roughly half-normal because the distribution of the angles is approximately normal. Note

³The mode here is calculated as the angle associated with the highest bar in the histogram of angles.

that there are many cut line angles close to the mode. If we think of there being a main axis of disagreement around a line with an angle of 20 degrees, that axis of disagreement can be said to explain most of the variation in choices on most of the items. For only 15 of 133 items are the cut line angles closer to 20 than to -70. This is why adding a second dimension only improves our classification of the responses by 1 percentage point in this dataset. Even when we allow for two dimensions, we see that a single ideological dimension explains most of the variation in responses across questions. In the Appendix, we provide more details on these analyses and show similar results for the other data sets analyzed in our original paper.

Discussion and Conclusion

We appreciate the theoretical issues raised by B&L. We agree with them that there could be potential distributions of policy views that are not explained by the parsimonious model in Fowler et al. (2023). Their point that there could be multiple Conversionian groups, which were not well captured by our original model, is particularly valuable. However, while B&L raise valuable interpretational concerns, their empirical evidence does not refute our findings about the prevalence of moderates in the mass public.

In this paper, we have re-examined the robustness of the analyses and conclusions in our original publication to the interpretational questions raised by B&L. Specifically, we have investigated whether their critiques change any of our original substantive conclusions. Based on extensions of our original model and new analyses that speak to the validity and reliability of our measures, we have shown that our initial findings about the prevalence and importance of moderates in the electorate are sound and robust to B&L's critique.

We also provide several new pieces of evidence. First, we briefly summarized the evidence presented by B&L and explained why it does not undermine our substantive conclusions or the usefulness of our model. Directly addressing B&L's concern that one Conversionian category may be insufficient, we analyzed an extension of our model that allows for two groups of

Conversionians. When we implement this extension, we continue to find that most, but not all, Americans have views that are well summarized by a single ideological dimension, and ideological moderates are especially consequential for electoral selection and accountability.

Next, we provided additional evidence that our classifications make sense based on the ways we expect ideologues and moderates to behave. Using data not included in the classification model, we showed that moderates are far more likely to place themselves near the median of a continuous policy scale, while those we identify as extremists place themselves away from the median. Interestingly, those we classify as Conversionian look different still.

We also compare our model to the most obvious alternative, that public opinion is multi-dimensional. We conducted additional analyses to better assess whether our data are usefully summarized by a one-dimensional model, and the results largely corroborate the value of this approach. A one-dimensional model accurately predicts approximately 75 percent of policy positions, and the addition of a second dimension only minimally improves fit. Furthermore, if we estimate a two-dimensional model, we find that most of the indifference lines are approximately parallel to one another, consistent with a model in which public opinion is approximately one-dimensional.

Our findings on public moderation are corroborated by other, independent evidence. For example, Hernandez (2026) asks a large battery of detailed questions about abortion policy and finds that public opinion is more nuanced, more moderate, and less polarized than standard questions suggest. Using a continuous policy question, Simonovits and Payson (2023) find that most Americans would prefer a local minimum wage that is higher than the federal minimum wage but lower than the current proposals of Democratic politicians and progressive city councilors. Analyzing continuous questions across 8 different issues, Fowler (2026) finds that Americans are more ideologically moderate and more constrained across issues than analyses of standard survey questions typically suggest. Of course, preferences are not perfectly correlated across issues, and some Americans do hold extreme views, but the weight of the evidence suggests that many Americans have policy preferences that are

close to the mean of the population and more moderate than the typical positions taken by candidates and elected officials.

Perhaps the most prominent evidence in support of public extremism comes from Brookman (2016). However, even in that study’s data, there are typically few Americans far from the mean or median of the population. Given the way these particular questions were posed, the American public seems to lean left on guns and taxes, and it leans right on unions and affirmative action. But if we define the middle as the mean or median of the sample, most respondents are close to the middle.

Overall, the weight of the evidence suggests that for a large majority of Americans, their policy preferences are usefully summarized by a single ideological dimension, and among those individuals, there are many with relatively moderate preferences. Although we find a meaningful minority of Americans whose views are less usefully summarized by a single ideological dimension, they still largely vote and behave as if they are Downsian moderates. Therefore, although standard scaling approaches that assume everyone is a Downsian might simplify a segment of the electorate with idiosyncratic views, they are still likely to do a good job predicting political behavior and characterizing electoral coalitions. While our findings largely support the continued use of one-dimensional models, we also hope that our methods will prove useful to future researchers who want to extend these models or further probe their usefulness. For example, mixture models like ours may be particularly useful to researchers studying preferences over new or more local issues or samples with higher shares of inattentive respondents.

We know this exchange will not be the last word on this topic, and we encourage future researchers to improve upon our methods and find better ways to summarize public opinion. We welcome increased attention to the extent of ideological moderation in the electorate — and the importance of moderates and others for political outcomes. We urge researchers to clearly define what they mean when they discuss ideological moderation. Our notion of moderation has to do with proximity to the mean of the population on a single ideological

dimension. Researchers who estimate multiple dimensions of public opinion will have to use another notion, and they should consider the possibility that someone might appear, by chance, to be far from the mean on one of the dimensions even if they are ideologically moderate. We look forward to continued research on this important topic.

Data Availability Statement

Research documentation and data that support the findings of this study are openly available at the American Political Science Review Dataverse: <https://doi.org/10.7910/DVN/FP965C>.

Acknowledgments

We thank David Broockman and Ben Lauderdale for this productive scholarly exchange.

Conflict of Interest

The authors declare no ethical issues or conflicts of interest in this research.

Ethical Standards

The authors affirm this research did not involve human participants.

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Supplementary Appendix for “Assessing Moderation and Multidimensionality with Mixture Models”

Results with Two Conversion Groups

Figures S1 through S6 and Tables S1 through S3 replicate all analyses from our original paper but using estimates from an extension of our model with two Conversion groups. With the exception of Table S1, we pool the two Conversion groups together for these analyses.

The results are strikingly similar to those in our original paper. In this extension, more respondents are classified as Conversion, but we still classify a large majority as Downsonian. We continue to find that those we classify as Downsonian moderates and Conversionians are particularly willing to change the party they support in elections and particularly responsive to the ideologies and experience levels of candidates.

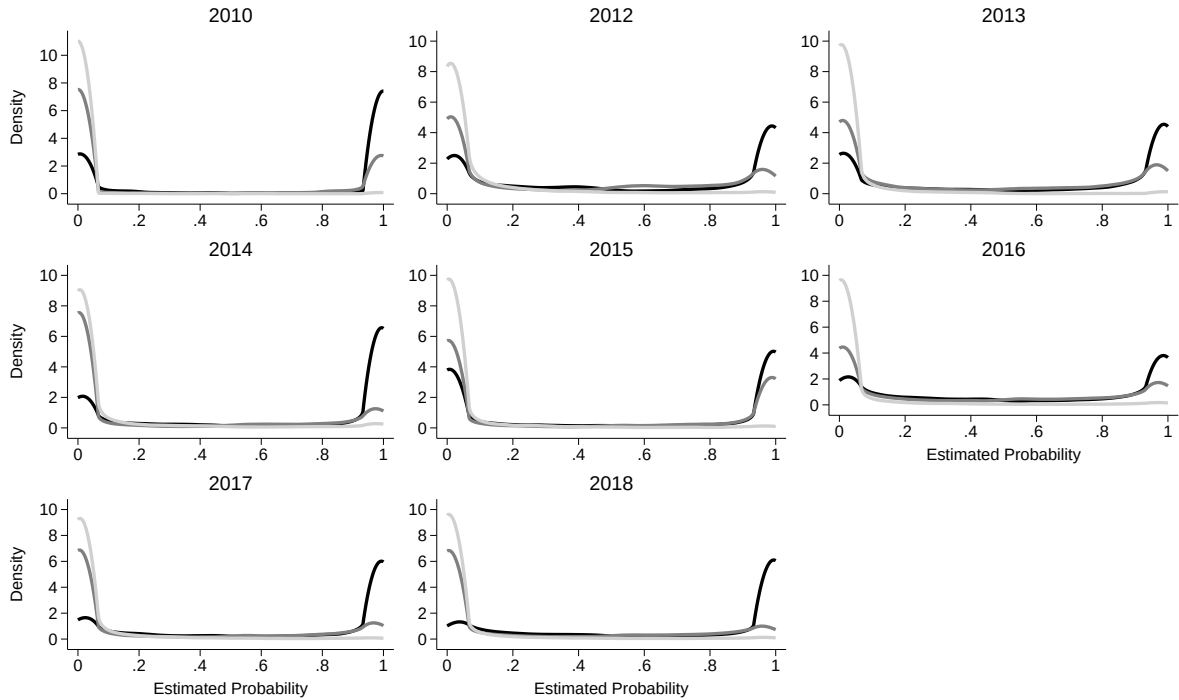


Figure S1: The figure shows kernel density plots (bandwidth = .03) of estimated probabilities that each respondent is a Downsonian (black), Conversionian (dark gray), and inattentive (light gray) type.

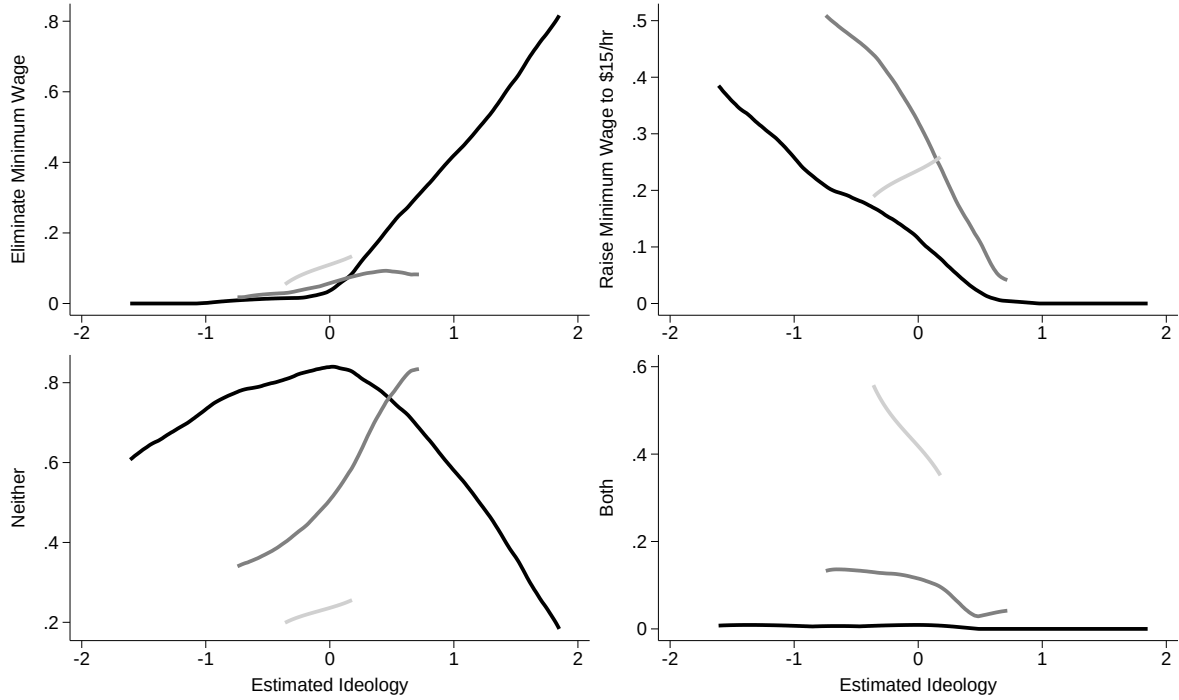


Figure S2: Kernel regressions (bandwidth = .03) of different positions on two minimum wage questions in the 2010 CCES module by a posteriori ideology. Separate plots are shown for Downsian (black), Conversionian (dark gray), and inattentive (light gray) respondents. The top-left panel shows support for eliminating the minimum wage and not raising it to 15 dollars. The top-right panel shows support for an increase to 15 dollars and not eliminating it. The bottom-left panel shows support for neither reform, and the bottom-right panel shows support for both reforms.

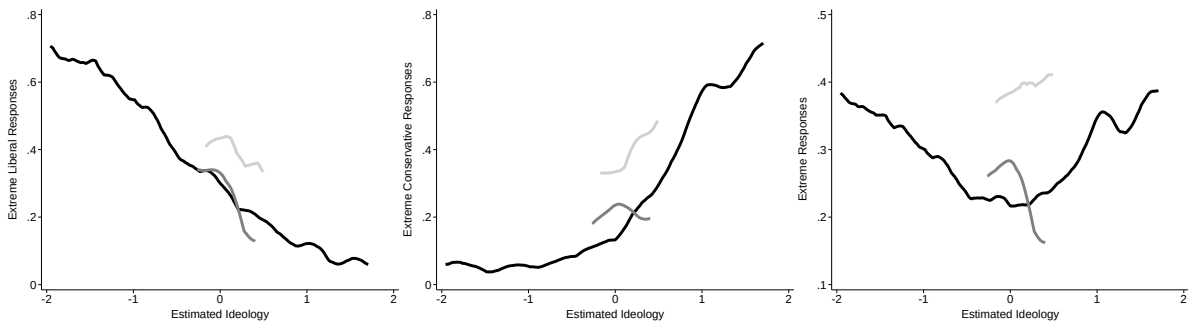


Figure S3: Kernel regressions (bandwidth = .01) of extreme policy positions by estimated ideology using out-of-sample questions. Separate plots are shown for Downsian (black), Conversionian (dark gray), and inattentive (light gray) types. Extreme positions are defined as responses to binary policy questions that are supported by less than 35 percent of respondents. The left panel examines the proportion of extreme liberal responses when such a response is possible. The center panel shows the analogous proportion of extreme conservative responses. The right panel shows the average of these two proportions.

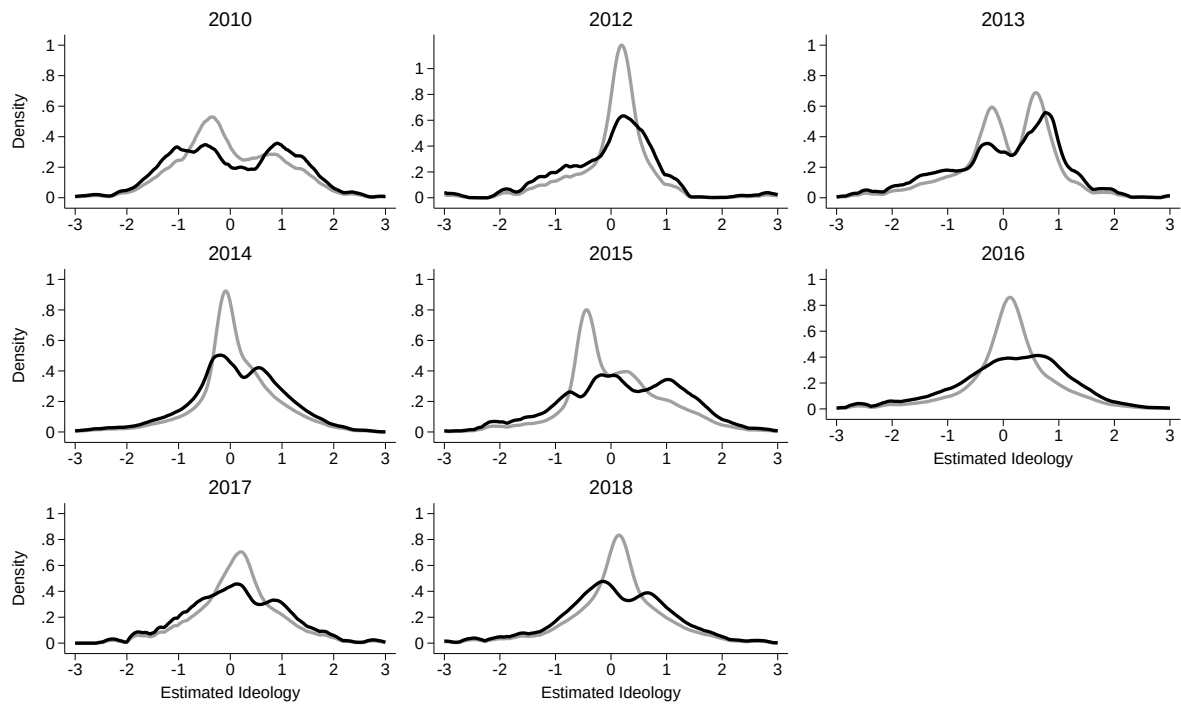


Figure S4: Distributions of Ideology for All Respondents and for Downsians

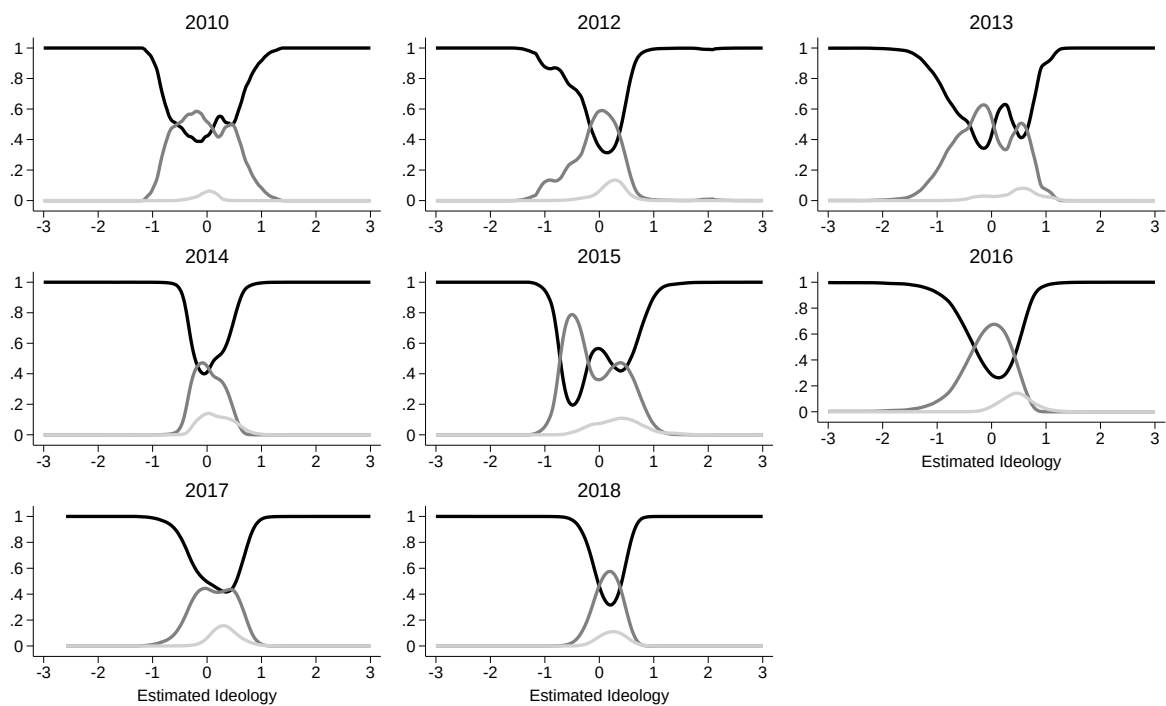


Figure S5: Respondent Type Probabilities by Estimated Ideology

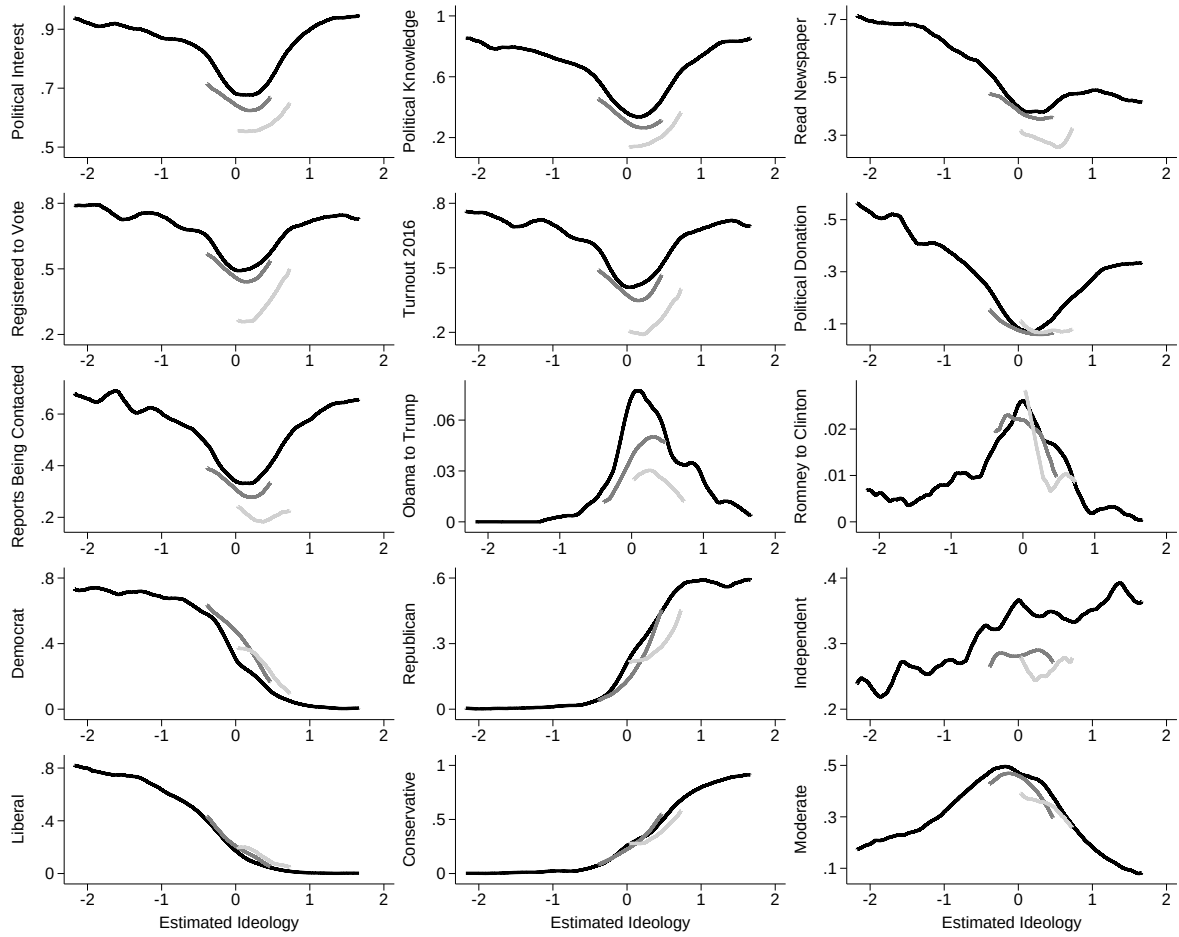


Figure S6: Kernel regressions (bandwidth = .1) of electoral attitudes and behaviors across estimated ideologies for Downsian (black), Conversionian (dark gray), and inattentive (light gray) respondents in the 2016 CCES.

Table S1: Issue Profiles across Categories in 2016, 2017, and 2018 CCES. The table shows the share of respondents in each category giving the conservative response to each question.

Item	Overall	Downsian:			Conversion 1	Conversion 2	Inattentive
		Conservative	Moderate	Liberal			
Three Strikes & Out Prison Sent.	.85	.95	.93	.70	.94	.80	.42
Buy & Hire American	.75	.96	.90	.48	.46	.92	.75
Strictly Enforce Imm. Laws	.53	.96	.67	.03	.10	.78	.70
Ban Late-Term Abortion	.65	.92	.69	.28	.85	.54	.41
Don't Publish Gun Owner Names	.59	.83	.67	.41	.58	.47	.47
Don't Use Medicaid for Abortion	.58	.94	.60	.08	.93	.38	.51
Repeal ACA	.50	.95	.52	.02	.57	.44	.50
Eliminate Income Tax	.47	.69	.44	.20	.51	.60	.48
Environmental Enforcement	.43	.97	.46	.01	.26	.40	.58
Withdraw Paris Climate Agreement	.41	.98	.31	.00	.35	.38	.47
Same-Sex Marriage	.39	.78	.30	.04	.55	.15	.61
Assault Weapon Ban	.36	.78	.34	.06	.27	.25	.64
Don't Increase Minimum Wage	.31	.79	.28	.03	.19	.14	.39
Make Abortions Illegal	.18	.29	.06	.02	.35	.10	.48
Body Cameras on Police	.13	.26	.11	.03	.07	.10	.46

Table S2: Average Estimates across Data Sources

Data Source	N	E[Downsian]	E[Conversion]	E[Inattentive]
CCES 2010 (module)	1,300	.693	.298	.009
CCES 2012	54,535	.579	.350	.071
CCES 2013	16,400	.595	.368	.037
CCES 2014	56,200	.697	.226	.077
CCES 2015	14,250	.552	.403	.044
CCES 2016	64,600	.569	.377	.053
CCES 2017	18,200	.696	.252	.052
CCES 2018	60,000	.709	.241	.050
Pooled	285,485	.635	.306	.059

Table S3: How Do Different Types Respond to Candidate Characteristics?

<i>DV = House Vote (Dem = 1, Rep = 0, Abstain/Other = .5)</i>									
	X = Ideological Midpoint			X = Incumbency			X = Experience		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
X*Moderate	.058 (.013)	.057 (.014)	.058 (.014)	.087 (.009)	.089 (.009)	.089 (.009)	.083 (.009)	.086 (.009)	.085 (.009)
X*Conversionian	.062 (.012)	.061 (.012)	.064 (.012)	.087 (.008)	.083 (.008)	.078 (.008)	.082 (.008)	.079 (.008)	.074 (.008)
X*Inattentive	.035 (.023)	.031 (.022)	.034 (.022)	.061 (.014)	.052 (.013)	.048 (.013)	.056 (.014)	.048 (.014)	.043 (.014)
X*Conservative	.018 (.014)	.016 (.015)	.019 (.015)	.012 (.012)	.018 (.012)	.018 (.012)	.004 (.012)	.011 (.013)	.011 (.013)
X	.023 (.009)	-.015 (.010)		.062 (.007)	-.012 (.011)		.067 (.007)	-.020 (.011)	
Moderate	-.395 (.007)	-.385 (.008)	-.385 (.008)	-.377 (.006)	-.375 (.006)	-.373 (.006)	-.378 (.006)	-.375 (.006)	-.373 (.006)
Conversionian	-.348 (.007)	-.340 (.007)	-.341 (.007)	-.329 (.005)	-.325 (.006)	-.324 (.005)	-.329 (.006)	-.325 (.006)	-.324 (.006)
Inattentive	-.379 (.013)	-.369 (.012)	-.373 (.012)	-.360 (.009)	-.354 (.009)	-.356 (.009)	-.359 (.009)	-.354 (.009)	-.354 (.009)
Conservative	-.738 (.008)	-.723 (.008)	-.723 (.008)	-.690 (.007)	-.684 (.007)	-.683 (.007)	-.687 (.008)	-.681 (.008)	-.680 (.008)
Year FEs	✓	✓		✓	✓		✓	✓	
District FEs		✓			✓			✓	
District-Year FEs			✓			✓			✓
Observations	159,006	159,006	159,006	233,445	233,445	233,445	233,445	233,445	233,445

District-clustered standard errors in parentheses. Liberals are the omitted category.

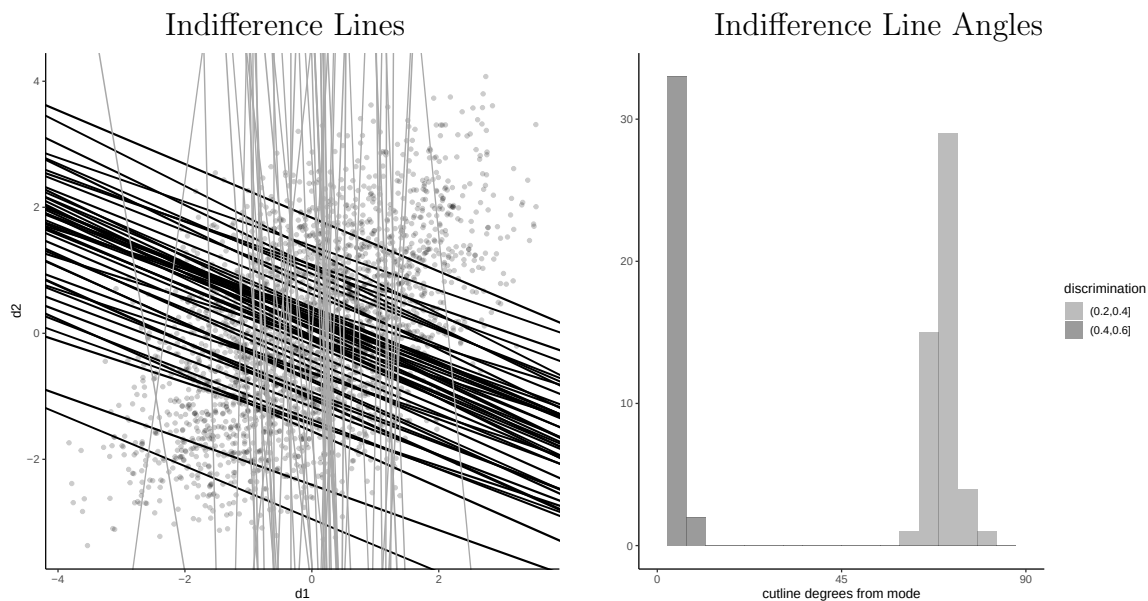
Additional Results on Dimensionality

In Figure D, we fit a two-dimensional model to responses from the 2010 CCES Module, plot the indifference lines, and show why a one-dimensional model so usefully summarizes policy preferences in this data set. Here, we provide more details on this analysis and show results for more data sets.

We start by showing how these results look when applied to B&L’s data simulated from a two-dimensional data generating process. The left panel of Figure S7 shows the indifference lines (sometimes called *cut lines*) of items generated by B&L’s simulation. Following B&L, we use the R package *mirt* to estimate their model. The points in the left panel of Figure S7 are the ideal points of hypothetical respondents, and the lines are the estimated indifference lines for the 100 questions that B&L simulate. The thickness of the lines is proportional to their discrimination, measured as the change in the probability of responding affirmatively to a question as an ideal point moves across the indifference line. The shading of the lines captures whether or not the angle of their slope is more than 45 degrees different from the modal angle, with items close to the mode represented in black and items far from the mode represented in gray.

The modal angle in this analysis is approximately -23 degrees, although indifference lines that are approximately 90 degrees (for the second dimension) are equally common; the choice of mode is arbitrary but does not change conclusions. The cutlines are not orthogonal because B&L’s two simulated dimensions, by construction, are correlated. Lines near -23 are all about the same width, somewhat wider than the lines with 90-degree angles because B&L generate data where one dimension is less discriminating than the other.

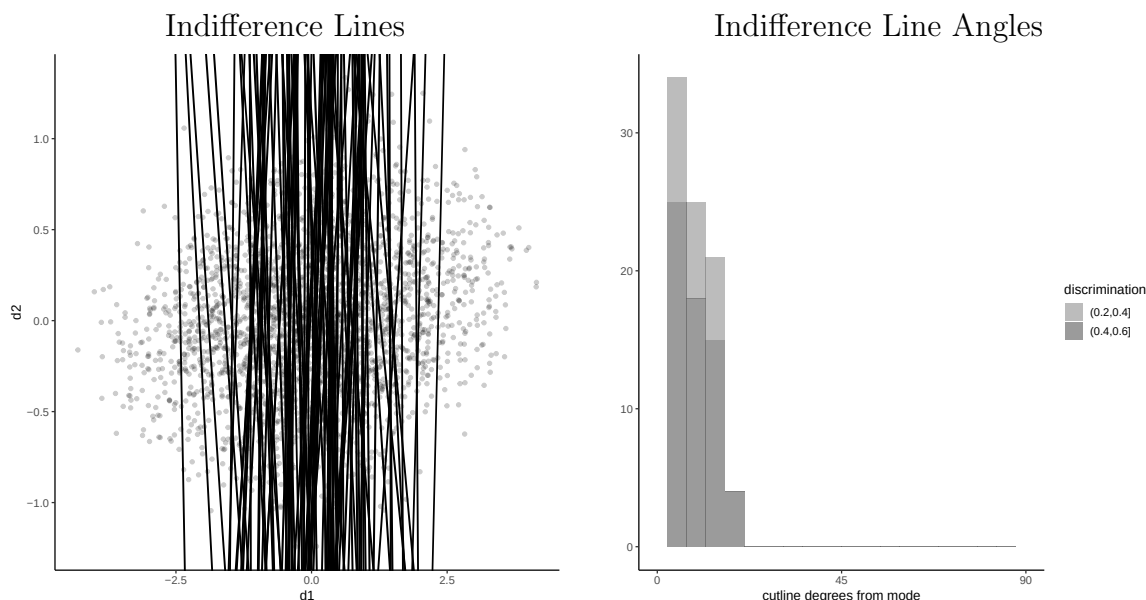
Figure S7: Analysis of Items in 2D Simulation



The right panel of Figure S7 shows a histogram of the absolute difference between the

angle of each indifference line and the modal angle.⁴ The bars of the histogram are shaded according to how discriminating the items are, with darker shades corresponding with more discrimination. Together, these two images illustrate a hallmark of data that is two dimensional: there are two modal values of the angle of the cutting line, each represents a substantial fraction of the items, and items in the second dimension have non-negligible discrimination. In a case like this, we could interpret the meaning of the second dimension by classifying the substantive differences between items with a roughly 90 degree indifference line angle and items with a -23 degree indifference line angle.

Figure S8: Analysis of Items in 1D Simulation



In Figure S8, we analyze data from the modified version of B&L’s simulation where the data generating process is one-dimensional. Although the survey responses are simulated from one dimension, we again estimate a two-dimensional model to fit the one-dimensional data, and graph the results in the same way as above.

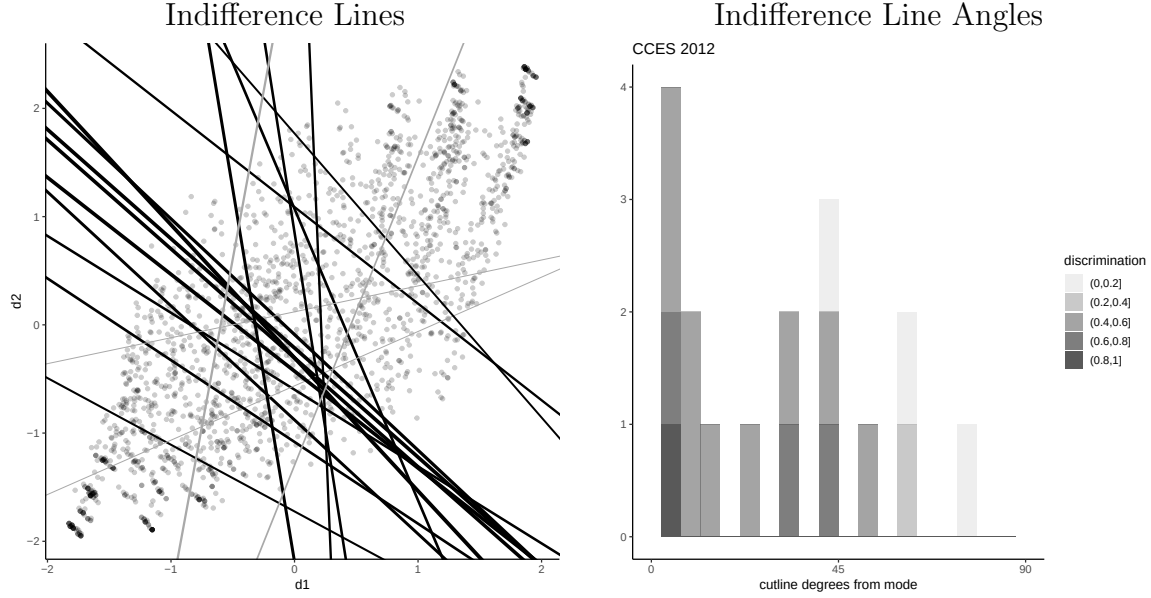
The difference between Figures S7 and S8 is clear. When the data are generated from a one-dimensional model, all indifference lines, even when estimated from a 2D model, have an angle that is close to the modal value of 90.

Figure S9 repeats the same estimation and graphical exercise on the 2012 CCES data from Fowler et al. (2023). The pattern of indifference lines is quite different from B&L’s two-dimensional simulation. Most of the indifference lines have an angle of -45 degrees. There are a few notable exceptions such as two lines with angles closer to 45 degrees—these are from two related immigration questions. Overall, however, there is a clear mode at -45 degrees with most of the questions appearing to divide respondents according to one dominant ideological dimension.

Another difference between Figure S9 and Figure S7 is that there is more variation in the discrimination of the items, as illustrated by the width of the lines. For the most

⁴The mode here is calculated as the angle associated with the highest bar in the histogram of angles.

Figure S9: Analysis of Items in 2012 CCES



discriminating item in 2012 (about support for the Affordable Care Act), a one-standard-deviation change in ideal points (from half a standard deviation below the indifference line to half a standard deviation above) is associated with an 81 percentage point change in response probability. For the median question, the corresponding statistic is 55. For 18 of the 21 items, the discrimination is at least 36 percentage points. The two questions producing anomalous angles near 45 degrees mentioned above are two of the least discriminating items, with corresponding statistics in the single digits. In other words, the few items that are poorly explained by the dominant dimension are not closely related to either dimension.

This pattern of a unimodal distribution of indifference angles with a few stragglers is mostly what we see when we look across our other data sets. Figure S10 shows the histograms of indifference line angles for each dataset. In almost every case, there is a clear mode and a few cut line angles that are far from this mode. In other words, patterns of responses from the real data sets look more like the one-dimensional simulation than the two-dimensional simulation.

These data sets have 21 to 34 different policy questions, but for the purposes of assessing dimensionality and making more nuanced distinctions, we might prefer even more items. This is why Figure D of the main text shows results for the subset of 2010 CCES respondents who answered 133 policy questions. With more items, we continue to see a unimodal distribution of indifference line angles, with most of them being approximately perpendicular to a single, dominant ideological dimension.

Figure S10: Indifference Line Angles in Moderates Data

